

Application — Real Product Analysis

Analysis Phase

The analysis phase focused on identifying inefficiencies in manual attendance, such as signature forgery and time loss. Researchers determined that a centralized, digital system was necessary to improve accuracy and streamline administrative processes. By documenting these requirements, the team established a clear scope for developing an effective e-attendance solution that aligns with institutional academic standards.

Mar'atutthahirah, M., Miftahulkhairah, M., & Alfito, R. B. (2024). Development of Student E-Attendance System using QR-Code and WhatsApp Gateway with The Iterative Model Based on Android. *Jurnal Teknologi Elektroika*, 21(2), 66–72.

<https://jurnal.poliupg.ac.id/index.php/JTE/article/view/5026>

Design Phase

The design phase translated requirements into a technical blueprint, utilizing the AppSheet platform for Android. Designers created flowcharts to map lecture initiation and QR-code scanning mechanics. By defining the system's architecture and user interface, the team created a clear, user-friendly roadmap for developers, ensuring the system would successfully integrate unique student QR identification with automated notification logic.

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Development Phase

Development involved building the Android-based application, integrating core features like the login interface and schedule forms. Crucially, developers implemented a WhatsApp Gateway API as a functional bridge, enabling the system to trigger real-time attendance notifications. This phase transformed design specifications into a tangible, functional product capable of handling automated data processing and user-side interactions.

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Testing Phase

In the testing phase, the team utilized black-box testing to verify that all features functioned according to design specifications. They rigorously checked components like the login process, QR-scanning, and WhatsApp messaging against expected outcomes. This quality assurance process ensured the application was stable, secure, and reliable, validating the system before its final deployment to end-users.

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Deployment Phase

Deployment involved releasing the application to the production environment for use by students and lecturers. Leveraging the cloud-based nature of the AppSheet platform, the system was distributed automatically across user devices. This allowed for efficient management, as updates to the database or logic were instantly synchronized, ensuring all stakeholders accessed the latest version without manual re-installation.

Atlassian. (n.d.). *Software deployment*. <https://www.atlassian.com/agile/software-development/software-deployment>

Maintenance Phase

Maintenance focuses on ensuring the system remains functional and relevant over time. This ongoing phase involves monitoring system performance, addressing reported bugs, and incorporating user feedback. By evaluating the system using tools like Likert scale surveys, developers identify areas for future improvements, ensuring the software adapts to evolving institutional needs and continues to provide accurate attendance documentation.

IBM. (2026). *What is the Software Development Lifecycle (SDLC)?*.
<https://www.ibm.com/think/topics/sdlc>